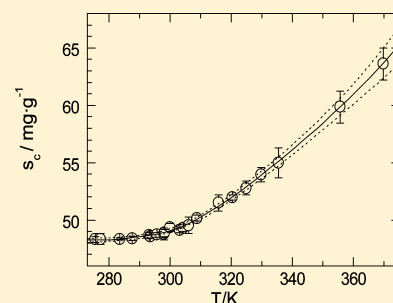


Solubility of Water in Octan-1-ol from (275 to 369) K

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ABSTRACT: The solubility of water in octan-1-ol was measured from 275.15 K to 369.15 K by coulometric Karl Fischer titration. The mass fraction solubility, s_w , at 298.15 K was found to be $(48.91 \pm 0.13) \text{ mg} \cdot \text{g}^{-1}$. Experiments indicated the existence of a pseudoequilibrium state when there was a significant difference in the temperature of the liquid and the headspace gas. The density of water-saturated octanol (WSO) was measured from 288.15 K to 318.15 K. The density at 298.15 K was determined to be $(831.7 \pm 1.7) \text{ kg} \cdot \text{m}^{-3}$ with a change in density of $-0.6 \text{ kg} \cdot \text{m}^{-3} \cdot \text{K}^{-1}$.



INTRODUCTION

The water and octan-1-ol system is a well-studied mixture as it has many utilitarian purposes. Since water and octan-1-ol are immiscible and exist as a biphasic system, it often serves as a model for the interface between blood or cytoplasm and the cell membrane. The sheer number of papers on the partition coefficient of drugs between water and octan-1-ol attests to this.^{1–3} However, as with many two-phase systems, there is a small but measurable amount of water that is soluble in the organic phase, and likewise there is some octan-1-ol in the aqueous phase.

The property of water being sparingly soluble in octan-1-ol has turned out to be extremely useful for Karl Fischer measurements. Specifically, water in octan-1-ol solutions are often used as calibration solutions for Karl Fischer titrations.⁴ These solutions are provided by many manufacturers in various concentrations, including the National Institute of Standards and Technology (NIST) standard reference material (SRM) 2890, water-saturated 1-octanol (WSO).^{4,5} The simplest calibration solution for Karl Fischer titrations would be a saturated system of water in octan-1-ol where there was both an aqueous and an organic phase to ensure saturation.

Several earlier studies report values of the solubility of water in octan-1-ol. The reported value of water in a saturated solution of octan-1-ol has varied over the years as methods and techniques improved for the measurement of water have improved. However, the differences in the reported values are larger than would be desired. One of the earliest measurement of the solubility of water in octan-1-ol was done by Leo and Hansch, where the reported value of water saturation was reported as “2.300 mol/L”; however, there was quite a bit of uncertainty in this value.⁶ Although this value of WSO was used as a reference for other solubilities, there was little utility in this value since there was no information on the density of the solution. Using the density of WSO determined in this study, the value reported by Leo and Hansch is equivalent to $49.9 \text{ mg} \cdot \text{g}^{-1}$.⁶ Further studies of WSO by Stephenson et al. give the temperature dependence of the solubility from 283 K to 363

K.⁷ The generally accepted value of WSO is that of Margolis as reported as part of the initial certification of NIST SRM 2890.^{4,5} In this same study Margolis also gives some temperature-dependent data for the water saturation level. While the NIST SRM has a certified value of water, it cannot by definition be “saturated” since it is not provided as a biphasic material. Furthermore, the Margolis data indicate a drop in the solubility of water at higher temperatures, for which there is no explanation.

This work concentrated on determining the solubility of water in octan-1-ol as a function of temperature using the best available methods. The anomalous decrease in water solubility above 303 K reported by Margolis has been investigated, and an explanation as to the cause is proposed.⁴

EXPERIMENTAL METHODS

Certain commercial equipment, instruments, or materials are identified in this paper in order to specify the experimental procedure adequately. Such identification is not intended to imply recommendation or endorsement by the National Institute of Standards and Technology, nor is it intended to imply that the materials or equipment identified are necessarily the best available for the purpose.

Anhydrous octan-1-ol was purchased from Sigma-Aldrich (lot 09474MH) with a mass fraction purity of 0.990. Water was obtained from a deionizing system (conductivity $\geq 18 \text{ M}\Omega$). The water content of the octan-1-ol was determined using a Metrohm 831 coulometric Karl Fischer system using a frit-less generator electrode with Hydranal Coulomat AG (Read al Hein, lot 9006A) as the working medium.

The coulometric Karl Fischer system was calibrated by injecting subsaturated water in octan-1-ol standard solutions that were prepared in the laboratory a few days prior to calibration. The octan-1-ol solution masses were determined by

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differential weighings on an analytical balance with a sensitivity of 0.01 mg. The calibration standards were prepared gravimetrically in oven-dried vials by the addition of deionized water to anhydrous octan-1-ol that had been placed over a 0.3 nm molecular sieve. The water content of the standard solutions ranged from approximately 20 mg·g⁻¹ to 47 mg·g⁻¹. A vial of dry octan-1-ol was used as the blank. All liquid transfers were performed under a dry nitrogen blanket that had been thoroughly purged to minimize uptake of water from the atmosphere. A dry magnetic stir bar was placed into each vial; the vials were then crimp-sealed and placed on a magnetic stirrer. Contents of the vials were stirred for several days at room temperature to ensure complete dissolution of the water, particularly in the more highly concentrated solutions.

Subsequent to the calibration of the Karl Fischer apparatus, the water content of WSO at various temperatures was measured. The primary batch of WSO was divided up into five different aliquots that were initially placed in water baths to bring the samples to one of several temperatures (i.e., 277 K, 288 K, 298.15 K, etc.). An additional bottle of WSO was left out on the bench at room temperature to serve as a reference point for the measurements. The method of temperature control was later expanded to include an incubator and an oven, both utilizing a large aluminum heat sink to hold the samples at higher temperatures than a water bath. The samples were allowed to equilibrate with intermittent shaking or stirring to help hasten the equilibrium. This was done for a minimum of 72 h before sampling and injection into the Karl Fischer system. The bottles at subambient temperature were left to equilibrate for much longer periods of time. All containers of WSO contained both an aqueous and an organic phase to ensure that the octanol was indeed saturated, and the two-phase system was maintained at all temperatures.

When the aliquots of the organic phase of WSO were sampled, care was taken to ensure that the organic phase was clear and that there were no water droplets in the organic phase, as confirmed by visual inspection. After the initial measurements were taken at a given temperature, the bottles were agitated and allowed to stand for another 24 h to 48 h and then remeasured to confirm that the sample was at equilibrium. For temperatures below 318 K, once sufficient replicates had been taken on a bottle at one temperature, the bottle would be transferred to a different temperature bath, and a second bottle would replace the first one at the first temperature. This was to ensure repeatability at a given temperature and to allow equilibrium to be approached from more than one direction.

Throughout the course of the experiment reference solutions of subsaturated water in octan-1-ol were measured to ensure a valid calibration. Temperature measurements below 333 K were made using a calibrated platinum resistance thermometer with a Tinsley Senator AC resistance bridge (SN 256694) to measure the thermometer resistance. The variation in the bath temperatures was less than 0.15 K. Above 333 K the temperature was measured with a calibrated mercury-in-glass thermometer, and the variation in temperature was on the order of ± 1.0 K.

Density measurements were made using an Anton-Paar 58 oscillating U-tube densimeter, which has a frequency display of six significant figures. Calibration of the densimeter was performed by measuring the vibrational periods of two materials with easily calculable densities. The first calibration point was air, where the density was determined based on the temperature, atmospheric pressure, and the relative humidity.

The second reference point was degassed deionized water, with the density calculated from the temperature from McCutcheon et al.⁸ Densities of the saturated solutions were measured from 288.15 K to 313.15 K, by sampling solutions that had come to equilibrium at the given temperatures.

RESULTS AND DISCUSSION

The calibration points for the Karl Fischer measurements provided a linear relationship between the water added by mass to the dry octan-1-ol and the values of the signal obtained from the Karl Fischer titration. A linear calibration curve of the form $r_{KF} = b \cdot w_{H_2O}$ was used to represent the data, where r_{KF} is the signal from the titration, w_{H_2O} is the gravimetrically determined mass fraction of water (in mg·g⁻¹), and b is the slope, with $b = 1.0029 \pm 0.0019$ as determined by linear regression, with the mean square error of the line as 0.13 mg·g⁻¹. Calibration data are given in Table 1. The uncertainty of the calibration was

Table 1. Calibration Points for the Coulometric Karl Fischer Titration^a

$w_{H_2O}/\text{mg}\cdot\text{g}^{-1}$	r_{KF}
20.255 $\pm$ 0.020	20.48 $\pm$ 0.24
24.387 $\pm$ 0.020	24.52 $\pm$ 0.26
29.676 $\pm$ 0.018	29.73 $\pm$ 0.12
33.283 $\pm$ 0.020	33.43 $\pm$ 0.20
37.467 $\pm$ 0.020	37.70 $\pm$ 0.34
40.222 $\pm$ 0.018	40.35 $\pm$ 0.36
43.378 $\pm$ 0.020	43.36 $\pm$ 0.32
46.691 $\pm$ 0.022	46.74 $\pm$ 0.19

^aGiven is the gravimetrically determined mass fractions of water in octan-1-ol, w_{H_2O} , and the signal or response factor, r_{KF} , from the Karl Fischer titration.

determined using a Working–Hotelling model.⁹ The uncertainty of the calibration near the value of WSO at room temperature was estimated to be ± 0.25 mg·g⁻¹ at the 95 % confidence interval. The calibration adjustment was applied to all of the measurements, and the uncertainties were used in later estimates of the overall uncertainty. The measured values of the reference solutions determined during the course of the experiment were all within the 95 % confidence interval.

The mass fraction solubility, s_c (as mg·g⁻¹), of water in octan-1-ol as a function of temperature is given in Table 2. Figure 1 shows the data collected in this study and how it compares with earlier studies. With the exception of the data from Leo and Hansch, the current results are higher than the previously reported values. The current values are 1.3 mg·g⁻¹ lower than the values reported by Leo and Hansch (reported as moles of water per liter of octan-1-ol, a difference of 2.6 %); however, they also report a range of values for WSO from 44.5 mg·g⁻¹ to 49.9 mg·g⁻¹, and they report that the higher value may be due to oversaturation.⁶ The data collected by Stephenson et al. is lower than the current work by 1.9 mg·g⁻¹ to 6.9 mg·g⁻¹, and there is a larger range of scatter than the current data collected.⁷ Lastly, analyzing the Margolis data shows that between 288.15 K and 303.15 K, the current data set is between 1.15 mg·g⁻¹ and 0.89 mg·g⁻¹ higher than the Margolis data.⁴ Furthermore, Figure 1 shows a second set of data taken during this study that, although correlating well with the Margolis data, was the result of a systematic bias and may be regarded as pseudosaturated and will be discussed subsequently.

Table 2. Mass Fraction Solubility, s_c , and Mole Fraction Solubility, s_x , of Water in Octan-1-ol as a Function of the Thermodynamic Temperature, T

T^a/K	$s_c^b/\text{mg}\cdot\text{g}^{-1}$	s_x
275.56 ± 0.36	48.32 ± 0.26	0.2684 ± 0.0014
277.31 ± 0.26	48.35 ± 0.50	0.2686 ± 0.0028
283.55 ± 0.02	48.36 ± 0.10	0.2687 ± 0.0010
287.78 ± 0.02	48.40 ± 0.16	0.2688 ± 0.0009
293.15 ± 0.12	48.67 ± 0.22	0.2700 ± 0.0012
293.47 ± 0.14	48.56 ± 0.15	0.2695 ± 0.0008
295.67 ± 0.22	48.71 ± 0.22	0.2701 ± 0.0012
298.13 ± 0.02	48.79 ± 0.50	0.2705 ± 0.0028
298.28 ± 0.02	48.88 ± 0.32	0.2709 ± 0.0018
299.99 ± 0.06	49.36 ± 0.33	0.2729 ± 0.0018
303.06 ± 0.06	49.12 ± 0.20	0.2719 ± 0.0011
304.23 ± 0.04	49.31 ± 0.63	0.2727 ± 0.0035
306.05 ± 0.16	49.55 ± 0.69	0.2737 ± 0.0038
308.81 ± 0.18	50.18 ± 0.26	0.2764 ± 0.0014
315.89 ± 0.22	51.51 ± 0.70	0.2819 ± 0.0038
320.48 ± 0.16	51.96 ± 0.22	0.2838 ± 0.0012
324.81 ± 0.32	52.77 ± 0.60	0.2871 ± 0.0033
329.81 ± 0.30	53.95 ± 0.61	0.2919 ± 0.0033
335.5 ± 1.0	55.0 ± 1.3	0.2961 ± 0.007
355.6 ± 1.6	59.9 ± 1.4	0.3152 ± 0.0074
369.7 ± 2.0	63.6 ± 1.4	0.3293 ± 0.0072

^aThe uncertainty for temperature was calculated as twice the standard deviation of replicate temperature measurements. ^bThe uncertainty of the solubility is represented by twice the experimental standard deviation since this value was larger than uncertainties calculated by propagation of error and Monte Carlo methods.

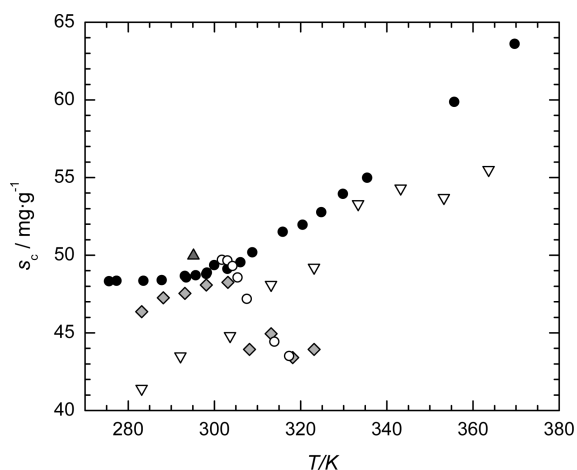


Figure 1. Mass fraction solubility of water, s_c , in octan-1-ol as a function of the thermodynamic temperature, T , from 275 K to 369 K, showing the best data from this study, ●, and from previous measurements in the literature (shown are data from Leo and Hansch,⁶ gray △; Stephenson et al.,⁷ ▽; and Margolis,⁴ gray ◇). Additionally shown are data from this work that represents a pseudosaturated state, ○.

To model the WSO data given in Table 2, two different polynomial equations were fit to the data to obtain the best empirical representation. Below 295.15 K the mass fraction solubility, s_c , of water (as $\text{mg}\cdot\text{g}^{-1}$), is well-represented by a quadratic equation of the form:

$$s_c = a + ct^2 \quad (1)$$

where $t = T - 298.15$, and T is the thermodynamic temperature. Above 295.15 K, the data set is well-represented by a fourth-order polynomial of the type:

$$s_c = a + bt + ct^2 + dt^3 + et^4 \quad (2)$$

Coefficients for eqs 1 and 2 are given in Table 3. The results of the fit for WSO from 273.15 K to 373.15 K are listed in Table 4. The original data along with the fit can be seen in Figure 2. The uncertainty of the fit is based on two techniques. The first uses the prediction interval calculated using the linear model prediction routine in the R programming language.¹⁰ The second estimation of error was determined using a Parametric Bootstrap Monte Carlo method (PBMC).^{11,12} Both methods were in good agreement below 323 K; however, PBMC gave a more realistic (larger) estimation of the uncertainty at higher temperatures. Thus the PBMC was used in the overall estimation of uncertainty; and these are the values given in Table 4 and displayed in Figure 2 as the dotted lines.

Since the solubility of water in octan-1-ol is a function of temperature, the thermodynamic parameters can be calculated from the mole fraction of water in octan-1-ol, $x_{\text{H}_2\text{O}}$ (values for $x_{\text{H}_2\text{O}}$ are given in Table 1). The solubility of water in octan-1-ol can be expressed as the equilibrium



where $\text{H}_2\text{O}_{(l)}$ is pure water and $\text{H}_2\text{O}_{(\text{oct})}$ is water dissolved in octan-1-ol. The standard state solubility, s_x° , can be calculated from the measured mole fraction solubilities s_x of water in octan-1-ol:

$$s_x^\circ = \frac{\gamma_{\text{H}_2\text{O}} \cdot s_x}{a_{\text{H}_2\text{O}(l)}} \quad (4)$$

Here, $\gamma_{\text{H}_2\text{O}}$ is the activity coefficient of the water in the octanol phase, and $a_{\text{H}_2\text{O}(l)}$ is the activity of the water in the aqueous phase, which by convention is taken to be unity.

The standard state for the solute is taken to be the hypothetical ideal solution at $x_{\text{H}_2\text{O}} = 1$, and the standard state of the solvent (octan-1-ol) is taken to be the pure solvent.

Considering the low concentrations of H_2O in the octan-1-ol solvent, it is assumed that the values of $\gamma_{\text{H}_2\text{O}}$ are equal to unity. Thus, the values of s_x° and s_x are equal, when the above assumptions are employed.

The measured values of s_x were fit using the model of Clarke and Glew with the following results for the reaction expressed in eq 3 at 298.15 K: $s_x = (0.2716 \pm 0.0006)$; $\Delta_r G^\circ = (3.231 \pm 0.006) \text{ kJ}\cdot\text{mol}^{-1}$; $\Delta_r H^\circ = (0.983 \pm 0.098) \text{ kJ}\cdot\text{mol}^{-1}$; and $\Delta_r C_p^\circ = (47 \pm 5) \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$.¹³ These results were obtained by fixing $(\partial \Delta_r C_p^\circ / \partial T)_p = 0$. If $(\partial \Delta_r C_p^\circ / \partial T)_p$ is made a part of the fit, the following values are obtained: $s_x = (0.2709 \pm 0.0004)$; $\Delta_r G^\circ = (3.238 \pm 0.004) \text{ kJ}\cdot\text{mol}^{-1}$; $\Delta_r H^\circ = (1.032 \pm 0.060) \text{ kJ}\cdot\text{mol}^{-1}$; $\Delta_r C_p^\circ = 65 \pm 7 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$, and $(\partial \Delta_r C_p^\circ / \partial T)_p = -(0.9 \pm 0.3) \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$. The uncertainties are equal to two estimated standard deviations of the mean and do not reflect any possible systematic errors in the measurements. Note that, although the value of $(\partial \Delta_r C_p^\circ / \partial T)_p$ is larger than its statistical uncertainty, it may have little real meaning. Furthermore, the fit is of the thermodynamic parameters is best in the region between 283 K and 313 K, and outside of this range the fit deviates from experimental values.

Table 3. Coefficients for Equations 1 and 2^a

	coefficient				
	$a/(\text{mg}\cdot\text{g}^{-1})$	$b/(\text{mg}\cdot\text{g}^{-1}\cdot\text{K}^{-1})$	$c/(\text{mg}\cdot\text{g}^{-1}\cdot\text{K}^{-2})$	$d/(\text{mg}\cdot\text{g}^{-1}\cdot\text{K}^{-3})$	$e/(\text{mg}\cdot\text{g}^{-1}\cdot\text{K}^{-4})$
eq 1	48.2536		$9.7652\cdot 10^{-4}$		
eq 2	50.640	$-2.5330\cdot 10^{-1}$	$9.3060\cdot 10^{-3}$	$-8.5290\cdot 10^{-5}$	$3.1653\cdot 10^{-7}$

^aThese equations give the mass fraction solubility, s_c , of water in octanol (as $\text{mg}\cdot\text{g}^{-1}$), as a function of the Celsius temperature, t , where $t = T - 273.15$. Below 295.15 K the data were modeled as $s_c = a + ct^2$ (eq 1), and above 295.15 K the data were modeled as $s_c = a + bt + ct^2 + dt^3 + et^4$ (eq 2).

Table 4. Smoothed Data for the Mass Fraction Solubility, s_c , of Water in Octan-1-ol as a Function of Temperature from the Fit of the Experimental Data, with the 95 % Confidence Interval As Determined via a Parametric Bootstrap Monte Carlo Simulation

T/K	$s_c/(\text{mg}\cdot\text{g}^{-1})$
273.15	48.25 ± 0.17
278.15	48.28 ± 0.16
283.15	48.35 ± 0.15
288.15	48.47 ± 0.14
293.15	48.65 ± 0.13
295.15	48.74 ± 0.13
298.15	48.91 ± 0.13
303.15	49.37 ± 0.14
308.15	49.99 ± 0.17
313.15	50.75 ± 0.20
318.15	51.61 ± 0.23
323.15	52.56 ± 0.27
328.15	53.57 ± 0.31
333.15	54.62 ± 0.36
338.15	55.72 ± 0.41
343.15	56.85 ± 0.48
348.15	58.02 ± 0.57
353.15	59.23 ± 0.67
358.15	60.49 ± 0.81
363.15	61.81 ± 0.99
368.15	63.2 ± 1.2
373.15	64.7 ± 1.5

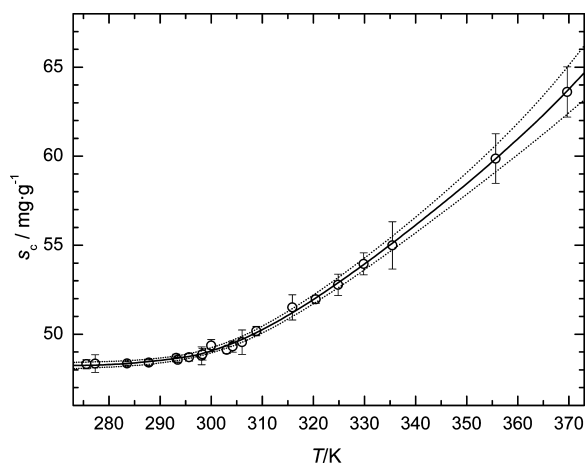


Figure 2. Mass fraction solubility, s_c , of water in octan-1-ol with the corresponding uncertainties for the data points. Given also is a smoothed fit of the data, —, and the 95 % confidence interval, ---, as calculated by parametric bootstrapping Monte Carlo.

An important point to note is that, during the course of the experiment, measurements on a nonthermally controlled aliquot of water and octan-1-ol were made periodically, with

the room temperature measured between 294 K and 295 K. The moisture in octan-1-ol under these conditions was measured at $(48.3 \pm 0.3) \text{ mg}\cdot\text{g}^{-1}$ as determined over several months. This value is consistently lower than the value determined at 295.15 K using a thermally controlled sample; this difference is statistically significant at a 95 % confidence level based on a t -test of the two means. Although the exact cause of this difference is unknown, it is an important point of reference since samples of WSO are often used at ambient conditions and this data will reflect many applications of the practical use of WSO.

As mentioned previously, a second set of data was taken during this study that has been classified as a pseudosaturated state. The pseudosaturated data was collected by allowing the headspace of the sample bottles to be above the level of the temperature-controlled bath so that the headspace was closer to ambient temperature. This was done to attempt to replicate what may have been done in previous measurements of WSO.⁴

From Figure 1, it can be seen that the water saturation begins to decrease above 303 K for water and octan-1-ol in this pseudosaturated state. This trend follows the observations originally reported by Margolis and is compatible with the Margolis data at the higher temperatures.⁴ The pseudosaturation levels were fairly repeatable, and the moisture values were generally consistent when the equilibrium point was approached from both higher and lower temperatures.

It is hypothesized that the dissimilarity between the equilibrated WSO and the pseudoequilibrium WSO system is a product of the difference of the temperature and pressure of the vapor phase. From basic thermodynamics, the chemical potential, μ , and by extension the solubility of a component in a system of immiscible liquids are directly related to the vapor pressure, P_i , of the component by the equation:

$$\mu = \mu_i^* + RT \ln \frac{P_i}{P_i^*} \quad (5)$$

where R , μ_i^* , and P_i^* are respectively the gas constant, the chemical potential, and standard state pressure of the pure component i , and if one assumes an ideal gas, the chemical potential can be expressed as a function of temperature.¹⁴ Thus, perturbations in the pressure (and ultimately temperature) of the vapor phase will cause a shift in the chemical potential and ultimately affect the solubility of components in the system. Clearly one desires to maintain the overall system such that the temperature of the vapor phase, T_g , is the same as that of the liquid phase, T_l . However, in the case of the pseudoequilibrium WSO system, if $T_g \neq T_l$, the chemical potentials of the octanol and water in the gas phase will be different than their chemical potentials for the case $T_g = T_l$. Since eq 5 applies to both the liquid and the vapor phases, and since these two phases are in equilibrium, changes in the vapor phase will ultimately create a new state of equilibrium in the liquid phase. This is the

Table 5. Density of WSO with Respect to Temperature^a

T/K	$\rho_{[\text{wso}]}(\text{expt})$ kg·m ⁻³	$\rho_{[\text{wso}]}(\text{fit})$ kg·m ⁻³	$\rho_{[\text{H}_2\text{O}]}^b$ kg·m ⁻³	$\rho_{[\text{oct}]}^c$ kg·m ⁻³	V_m^{ex} 10 ⁻⁶ m ³ ·mol ⁻¹
288.15	836.1 ± 0.5	836.6 ± 3.0	999.1	828.6	-0.154 ± 0.066
293.15	832.6 ± 0.5	833.2 ± 2.3	998.2	825.2	-0.137 ± 0.071
298.15	829.7 ± 0.5	829.7 ± 1.8	997.1	821.8	-0.124 ± 0.068
301.65	827.5 ± 0.6	827.3 ± 1.7	996.1	819.3	-0.117 ± 0.067
303.15		826.3 ± 1.5	995.7	818.3	-0.114 ± 0.066
305.65	826.2 ± 0.7	824.6 ± 1.6	994.9	816.5	-0.111 ± 0.067
308.15	824.3 ± 0.9	822.9 ± 1.6	994.1	814.8	-0.107 ± 0.066
310.15	820.6 ± 0.8	821.5 ± 1.8	993.4	813.3	-0.106 ± 0.067
313.15	820.4 ± 1.0	819.5 ± 2.1	992.2	811.2	-0.105 ± 0.069
318.15	814.9 ± 1.3	816.1 ± 2.7	990.3	807.6	-0.105 ± 0.073

^aGiven are the densities of WSO as determined from the experiment, $\rho_{[\text{wso}]}(\text{expt})$, densities from the fit of the data, $\rho_{[\text{wso}]}(\text{fit})$, density of pure water at the given temperatures, $\rho_{[\text{H}_2\text{O}]}$, density of pure octanol, $\rho_{[\text{oct}]}$, and the excess volume of the WSO system, $V_m^{\text{ex}} = (V_m - V_m^{\text{id}})$, where V_m is the experimentally determined molar volume and V_m^{id} is the ideal molar volume. Uncertainties for the fit are from a Working–Hotelling estimate of the uncertainty. ^bDensity of water from Handbook of Hydrology.⁸ ^cDensity of octan-1-ol from NIST/TRC Web Thermo Tables.¹⁵

pseudoequilibrium, and it will be different than the true equilibrium where all parts of the system are at the same temperature. Thus it is imperative that liquid and vapor phases be maintained at the same temperature to achieve the ideal equilibrium conditions.

The density of the organic phase of the water and octan-1-ol system was measured from 288.15 K to 318.15 K. The results are given in Table 5. The data were used to generate a linear fit of the density, ρ , as a function of temperature as $\rho_{\text{wso}} = -(0.684 \pm 0.083) \cdot t + (846.8 \pm 2)$, where $t = T - 298.15$, with the mean square error of the line as 2.7 kg·m⁻³. The data points along with the fit are shown in Figure 3, where the solid line is the fit

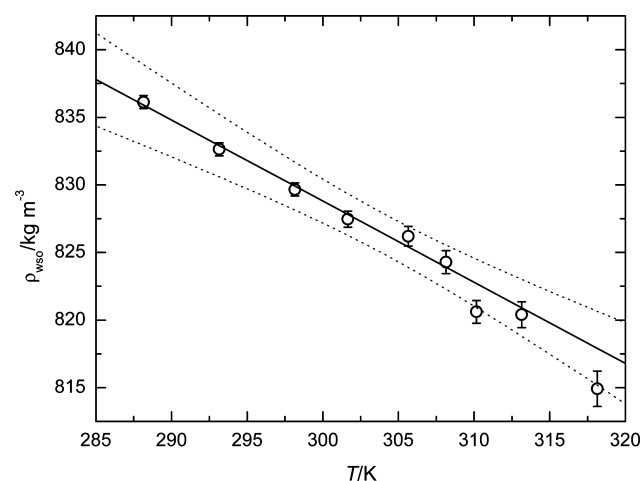


Figure 3. Density of WSO as a function of temperature showing curves representing best fit regression of the data, —, along with 95 % confidence intervals for the curves, ···, determined from a Working–Hotelling model.

of the data and the dotted lines show the uncertainties of the data. Table 5 also gives the density as determined from the fit. Confidence intervals of the fit were determined with the Working–Hotelling method and are reflected in the uncertainties listed in Table 5. Table 5 also gives the bulk densities of water and of octan-1-ol, $\rho_{\text{H}_2\text{O}}$ and ρ_{oct} respectively. The bulk densities and the respective mole fractions of water and octan-1-ol were used to determine the ideal molar volume, V_m^{id} , of the WSO at the various temperatures, where $V_m^{\text{id}} =$

$x_{\text{H}_2\text{O}} \cdot \rho_{\text{H}_2\text{O}} + x_{\text{oct}} \cdot \rho_{\text{oct}}$ with $x_{\text{H}_2\text{O}}$ and x_{oct} being the mole fractions of water and octan-1-ol, respectively. From the fit of the densities of the WSO, the experimental molar volume, V_m , was determined, which in turn was used to find the excess molar volumes, $V_m^{\text{ex}} = V_m - V_m^{\text{id}}$. The excess molar volumes are also given in Table 5. The negative values of V_m^{ex} indicate that there is a slight decrease in volume when the two components are mixed; however, the magnitude of V_m^{ex} is less than 0.1 % of the total molar volume, meaning that the solution behaves nearly to an ideal mixture.

CONCLUSION

The mass fraction solubility of water in octan-1-ol is relatively insensitive to temperature changes below 298.15 K. Above 298.15 K the solubility increases much more rapidly with respect to temperature. However, at higher temperatures the system is susceptible to changes in the partial pressures of the component solvents, leading to larger uncertainties at higher temperatures. This effect is magnified when there is a large temperature difference between the headspace and the liquid portion, resulting in a state of pseudoequilibrium.

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Notes

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